

Part I

SUMMARY of Antiderivatives:

Summary

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A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$, for all x in I .

If F is an antiderivative of f on an interval I , then the function f has a whole **family of antiderivatives**.

If a function G is an antiderivative of f on I , then $G(x) = F(x) + C$, for all x in I , for some constant C .

The family of of *all* antiderivatives of f is denoted by $\int f(x) dx$ and called **indefinite integral of f** . Therefore, if F is an antiderivative of f , $\int f(x) dx = F(x) + C$.

Basic Indefinite Integrals

- $\int k dx = kx + C$
- $\int \frac{1}{x} dx = \ln |x| + C$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
- $\int e^x dx = e^x + C$
- $\int a^x dx = \frac{a^x}{\ln(a)} + C$
- $\int \cos(x) dx = \sin(x) + C$
- $\int \sin(x) dx = -\cos(x) + C$
- $\int \sec^2(x) dx = \tan(x) + C$
- $\int \csc^2(x) dx = -\cot(x) + C$
- $\int \sec(x) \tan(x) dx = \sec(x) + C$
- $\int_C \csc(x) \cot(x) dx = -\csc(x) + C$
- $\int \frac{1}{x^2 + 1} dx = \arctan(x) + C$
- $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin(x) + C$

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 Find the most general antiderivative of the function

$$g(t) = e^{-2t} - 5 + 6\sqrt{t} - \frac{7}{t} + \frac{5}{1+t^2}$$

Problem 2

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Problem 2 Determine the following indefinite integrals.

(a)

$$\int (\sec^2(x) + 5) \, dx$$

(b)

$$\int (\sec^2(x) - \sec(x) \tan(x)) \, dx$$

(c)

$$\int \frac{x + 3x^5}{x^3} \, dx$$

(d)

$$\int \frac{1 + 2x}{1 + x^2} \, dx$$

(HINT: Think about the derivative of $\ln(f(x))$).

(e)

$$\int \frac{2 + x^2}{1 + x^2} \, dx$$

Problem 3

February 21, 2026

Problem 3

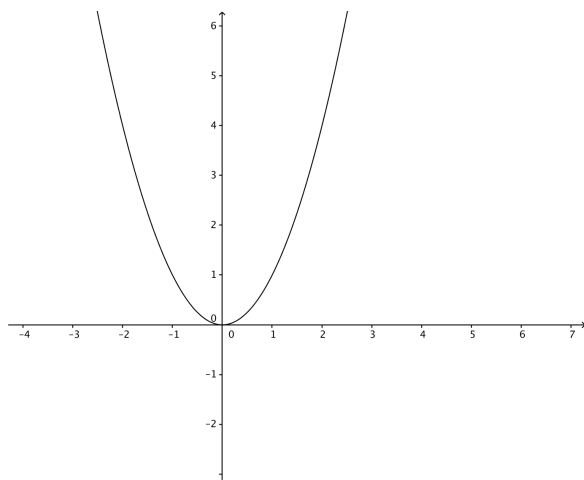
Problem 3 Assume that $f'(t) = 4t^3 + 2t$ and $f(3) = 5$. Find $f(t)$.

Problem 4

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Problem 4

Problem 4 The graph of a function f is the parabola given in the figure below.



- (a) Find a formula for $f(x)$.
- (b) Suppose F_1 is an anti-derivatives of f satisfying $F_1(0) = 2$. Sketch and label the graph of F_1 .
Sketch the graphs of three more antiderivatives of f .
- (c) Using the expression you found in part (a) and your sketches in part (b), find the algebraic representation of F_1 .
- (d) Suppose we're given $h(x) = (1/3)x^3 + 17$, what is $h'(x)$?
- (e) What is the relationship between f , F_1 , h , and h' ?

Problem 5

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Problem 5

Problem 5 *Given the acceleration function, the initial velocity and initial position of an object moving along a line, find the **position function**.*

$$a(t) = 4 \cos t, \quad v(0) = 2, \quad s(0) = 6.$$

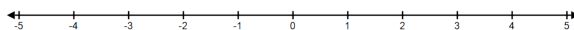
Problem 6

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Problem 6

Problem 6 Consider an object moving along a line with velocity $v(t) = \pi \sin(\pi t)$ on $[0, 2]$ and initial position $s(0) = 0$. Time is measured in seconds and velocity in m/s.

- (a) Determine the position function, $s(t)$, on $[0, 2]$.
- (b) Mark the position of the object at the time $t = 1$ on the line below.



- (c) Determine the average velocity, v_{av} , of the object during the interval $[0, 2]$.
- (d) Determine when the motion is in the positive direction.
- (e) At what time (or times) is the object farthest from the origin?

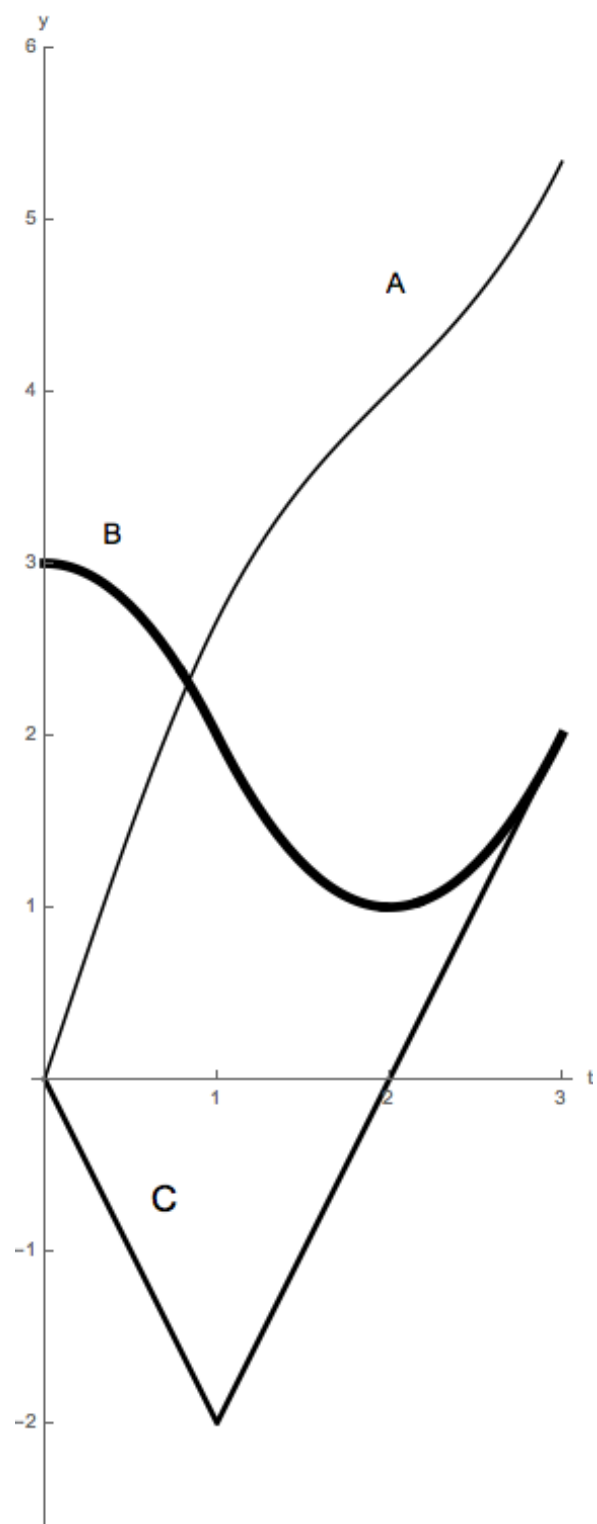
Problem 7

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Problem 7

Problem 7 Consider an object moving along a straight line. The graphs of acceleration function (in m/s^2), the velocity function and the position function of the object are given in the figure below.

Problem 7



Problem 7

Find the initial velocity, $v(0)$, and initial position, $s(0)$, of the object. (HINT: Start by determining which graph corresponds to each of $s(t)$, $v(t)$, and $a(t)$.)